# Generative AI Analysis Report

**Github Repo Link:** https://github.com/Shehzaad-S/generative-ai-applications-project

## Report Overview

This project explores conversational text generation using the Cornell Movie-Dialogs Corpus and a custom GPT-style decoder-only Transformer model. The objective was to determine whether a domain-specific tokenizer and carefully optimized Transformer architecture could learn meaningful conversational patterns from movie dialogue while remaining computationally feasible on modest GPU hardware. The study follows the autoregressive language modeling technique introduced by Generative Pretrained Transformers (GPT), in which the model learns to predict the next token in a sequence given all previous tokens (Radford et al., 2018), while leveraging the self-attention architecture proposed by Vaswani et al. (2017).

This work adopts a data-centric approach that places significant emphasis on corpus analysis, tokenizer design, conversational preprocessing, and controlled architectural experimentation. A custom Byte Pair Encoding (BPE) tokenizer was developed and experimentally evaluated against the GPT-2 tokenizer, resulting in improved token efficiency and a substantial reduction in vocabulary size. The resulting efficiency gains enabled progressively deeper Transformer architectures to be trained and evaluated within Google Colab resource constraints.

The project combines quantitative evaluation using validation loss, test loss, and perplexity with qualitative analysis of generated dialogue. Controlled experiments investigated the impact of increasing Transformer depth and regularization, culminating in a final eight-layer model that demonstrated improved conversational quality, stronger generalization performance, and more effective use of conversational boundary markers learned from the movie-dialogue corpus.

## Dataset Description

This project utilized the Cornell Movie-Dialogs Corpus, a large collection of fictional movie conversations extracted from film scripts and originally developed for the study of conversational behavior and discourse patterns (Danescu-Niculescu-Mizil & Lee, 2011). The raw corpus contains dialogue utterances together with metadata describing speakers, movies, and conversation structure. Unlike manually curated conversational datasets

designed specifically for natural language processing tasks, the Cornell corpus captures a wide variety of dramatic, emotional, comedic, and narrative interactions commonly found in film dialogue.

The original corpus was reconstructed into complete conversations consisting of 83,097 dialogue exchanges containing approximately 3.5 million words. Exploratory analysis revealed a median conversation length of 28 words and a mean conversation length of 42.31 words, indicating that the dataset is dominated by relatively short conversational interactions despite containing a small number of much longer exchanges. To preserve conversational structure during training, custom dialogue boundary tokens "<|eou|>" and conversation termination tokens "<|endofdialog|>" were introduced and retained throughout preprocessing.

A key design decision involved performing a tokenizer experiment rather than relying solely on an existing pretrained tokenizer. The corpus was analyzed using both the GPT-2 tokenizer and a custom Byte Pair Encoding (BPE) tokenizer trained directly on the movie-dialogue corpus. BPE tokenization is a widely adopted subword segmentation strategy that helps balance vocabulary size and token efficiency by representing text using frequently occurring subword units (Sennrich et al., 2016). Experimental results showed that the custom tokenizer reduced the vocabulary from 50,257 tokens to 16,384 tokens while maintaining coverage of conversational language patterns present in the corpus.

**Dataset Link:** https://www.cs.cornell.edu/~cristian/research#paper-chameleons-in-imagined

*(Click on **Legacy Data** to download)*

## Model Design and Training Approach

The model was implemented as a GPT-style decoder-only Transformer using the self-attention architecture introduced by Vaswani et al. (2017) and the autoregressive language modeling paradigm popularized by Generative Pretrained Transformers (Radford et al., 2018). The architecture consisted of learned token embeddings, learned positional embeddings, modern causal multi-head self-attention using PyTorch's optimized scaled_dot_product_attention() operation, feed-forward networks with GELU activations, residual connections, and pre-layer normalization. Causal masking was applied during attention computation to ensure that each token could attend only to previous tokens, preserving the next-token prediction objective.

Several design decisions were informed by exploratory analysis of the corpus. Conversation statistics showed that 95% of conversations contained fewer than 162 tokens, leading to the selection of a context length of 256 tokens. A conversation-aware train/validation/test split was implemented using "<|endofdialog|>" markers to ensure that complete conversations remained within a single data partition and to prevent dialogue leakage between training and evaluation datasets.

A custom BPE tokenizer was selected after experimental comparison with the GPT-2 tokenizer. The custom tokenizer reduced the vocabulary size from 50,257 tokens to 16,384 tokens while also reducing the overall number of tokens required to represent the corpus. This reduction significantly decreased embedding overhead and enabled larger Transformer architectures to be trained within the available computational budget.

Training was performed using AdamW optimization, mixed precision training (AMP), gradient clipping, and a batch size of 64. These techniques were selected to improve computational efficiency and training stability while operating within Google Colab GPU constraints. A baseline model consisting of 4 Transformer layers was first established, followed by controlled architectural experiments that increased model depth and dropout. The final model used 8 Transformer layers, an embedding dimension of 256, four attention heads, a feed-forward dimension of 1024, and a dropout rate of 0.20.

Experimental results demonstrated that increasing model depth consistently improved validation performance. The baseline trained model achieved a best validation loss of 4.4881, while subsequent experiments reduced validation loss from an **untrained** model loss of > 9. These findings suggest that increased depth and additional regularization improved the model's ability to learn conversational structure while reducing overfitting on the movie-dialogue corpus.

## Output Evaluation and Interpretation

Model performance was evaluated using both quantitative language modeling metrics and qualitative inspection of generated dialogue. The final model achieved a best validation loss of 4.2336, a test loss of 4.2925, and a perplexity of 73.15. The close agreement between validation and test loss suggests that the model generalized reasonably well to previously unseen conversations while avoiding severe overfitting.

Qualitative evaluation revealed that the model successfully learned several important characteristics of conversational dialogue. Generated examples consistently demonstrated turn-taking behavior, effective use of the custom "<|eou|>" and

“<|endofdialog|>” boundary markers, and the ability to sustain multi-turn interactions. The strongest outputs exhibited coherent conversational flow, emotionally aware responses, and narrative exchanges that reflected the dramatic and interpersonal style commonly found within movie dialogue.

Output quality improved noticeably as model depth increased. Early baseline generations were often short, generic, or occasionally incoherent, whereas deeper Transformer configurations produced longer and more contextually appropriate exchanges. The final eight-layer model generated plausible responses across greeting, emotional, and narrative prompts, indicating that it had learned meaningful conversational structure rather than simply memorizing short response patterns.

Despite these strengths, several limitations remained. The model occasionally produced repetitive responses, lost coherence during longer conversations, and sometimes generated statements with weak logical connections between turns. Additionally, evaluation using an out of domain factual question demonstrated that the model struggled to answer knowledge based prompts and instead produced conversational responses unrelated to the requested information. This behavior highlights the distinction between learning dialogue patterns and acquiring factual world knowledge.

Overall, the results suggest that the combination of a custom tokenizer, conversation-aware preprocessing, deeper Transformer architectures, and additional regularization enabled the model to generate coherent movie-style dialogue while maintaining reasonable generalization performance. Although the generated outputs remain less sophisticated than those produced by modern large language models trained on internet-scale corpora, the final model successfully achieved the project's objective of learning conversational structure from a relatively narrow dialogue dataset.

## Ethical Considerations and Responsible Use

One ethical concern associated with generative language models is their tendency to produce outputs that appear plausible despite lacking factual understanding. Because text generation is fundamentally probabilistic, generated responses may be misleading, inaccurate, or inconsistent while still appearing coherent to users. This project demonstrated this limitation during qualitative evaluation, where the model generated conversational responses to factual questions instead of providing correct answers.

A second consideration relates to bias inherited from the training data. Bender et al. (2021) note that language models frequently absorb and reproduce patterns, stereotypes, and

biases present within the corpora used during training. In this project, the model was trained exclusively on movie dialogue, meaning that generated responses may reflect the social assumptions, dramatic themes, character archetypes, and interpersonal biases commonly found within film scripts. Because these patterns are learned statistically rather than understood conceptually, the model has no inherent ability to recognize or correct problematic outputs.

Additionally, the model contains no content moderation or safety guardrails, meaning that inappropriate, misleading, or biased responses could potentially be generated under certain prompts. For these reasons, conversational language models should be viewed as generative tools rather than authoritative sources of information, and appropriate transparency and human oversight should accompany their deployment.

## Limitations and Future Improvements

Despite achieving encouraging results, several limitations remain. The model was trained exclusively on the Cornell Movie-Dialogs Corpus, which represents a relatively narrow domain and primarily exposes the model to scripted movie conversations. As observed during qualitative evaluation, the model demonstrated conversational fluency but struggled with factual question-answering, long-range coherence, and maintaining logical consistency across extended interactions. Additionally, computational constraints limited experimentation with larger architectures, longer context lengths, and more extensive hyperparameter optimization.

Several realistic improvements could be explored in future work. Expanding training data beyond movie dialogue would likely improve linguistic diversity, reduce domain-specific biases, and strengthen generalization to everyday conversations. Additional architectural experiments involving larger context windows, alternative attention mechanisms, and more systematic regularization strategies may further improve performance. From a deployment perspective, introducing safety guardrails, content filtering, and factual grounding mechanisms would help mitigate inappropriate or misleading outputs. Finally, the trained model could be integrated into an interactive application or chatbot interface to support more comprehensive evaluation of user experience and conversational quality.

## References


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